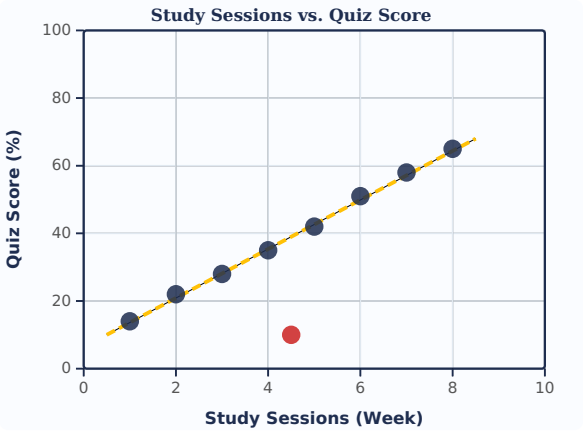
📝 Unit 7 Homework — Data Analysis — ANSWER KEY

Complete independently · NC.8.SP.1–4 · Show all work

Remember: Describe scatterplots using all four features — Direction, Strength, Form, and Outliers.

Question 1



Looking at the **non-outlier** points, what correlation is shown?

- ○ Negative correlation
- ✓ **Positive correlation**
- ○ No correlation
- ○ Perfect correlation

Question 2



This scatterplot shows athletes' skill level as weeks without practice increase. Describe the full correlation.

- ○ Weak positive correlation — practice doesn't matter
- ✓ **Strong negative linear correlation — skill drops significantly as inactivity increases**
- ○ No correlation — the points are scattered
- ○ Moderate positive — skill improves with rest

Question 3

In a scatterplot for "temperature vs. hot coffee sales," which type of correlation would you expect?

- ○ Strong positive
- ✓ **Strong negative**

✓ **Strong negative**

○ No correlation

○ Perfect positive

Question 4

A scatterplot shows "number of hours of daylight vs. hours of sleep." Which correlation is most likely?

○ Strong positive

✓ **Weak to no correlation**

○ Strong negative

○ Perfect negative

Question 5

Student scores a 95 on a test after studying 8 hours. The trend line predicts 87 points at 8 hours. This student's score above the line makes it:

✓ **A point above the trend line (the line under-predicts this student)**

○ An outlier below the trend line

○ The y-intercept

○ The maximum score

Question 6

Ice cream sales and pool visits both increase in summer. A student says "ice cream causes people to visit pools." What concept explains the real relationship?

○ Causation — ice cream makes people want to swim

✓ **Correlation does not imply causation — both are caused by hot weather**

○ The data must be wrong

○ There is no correlation

Question 7

Which pair would make a scatter plot with no pattern (no correlation)?

○ Hours studied vs. test score

○ Height vs. weight

✓ **Shoe size vs. IQ score**

○ Exercise vs. weight loss

Question 8

A trend line is described as "strong and positive." What does each word mean?

○ Strong = steep slope · Positive = y is positive

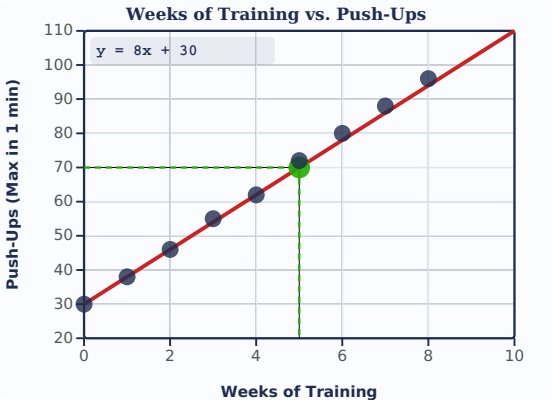
✓ **Strong = points cluster tightly near the line · Positive = trend goes upward**

○ Strong = many data points · Positive = all y-values are positive

○ Strong = the line is long · Positive = x values increase

Interpret in context: slope = "per unit increase in x" · y-intercept = "when x = 0"

Question 1



The model is y = 8x + 30 (x = training weeks, y = max push-ups in 1 min). Find the green dot: how many push-ups after **5 weeks**?

- ○ 38 push-ups
- ✓ **70 push-ups**
- ○ 55 push-ups
- ○ 80 push-ups

Question 2

In y = 8x + 30 (training weeks vs. push-ups), interpret the slope of 8.

- ○ The athlete starts at 8 push-ups
- ✓ **For each additional week of training, push-up count increases by about 8**
- ○ Training takes 8 weeks total
- ○ The slope means 8 athletes were tested

Question 3

Interpret the y-intercept of 30 in the same model (y = 8x + 30).

- ○ 30 is the maximum push-up count
- ✓ **At 0 weeks of training, a person can do about 30 push-ups**
- ○ It takes 30 weeks to improve
- ○ 30 people were surveyed

Question 4

Temperature vs. electric heating bills: y = −2x + 120 (x = °F above 50, y = bill in $). What does slope −2 mean?

- ○ The bill increases $2 per degree above 50°F
- ✓ **For each degree above 50°F, the bill decreases by $2**
- ○ The starting bill is −$2
- ○ The bill is always negative

Question 5

Using y = −2x + 120 from Q4: what is the predicted bill at 15 degrees above 50°F?

- ✓ **$90**
- ○ $105

○ $150

○ $120

Question 6

Model: y = 4x + 8 (x = hours tutoring, y = points gained). The data was collected for x = 1 to x = 6. Is predicting y at x = 10 an example of interpolation or extrapolation?

○ Interpolation — within the data

✓ **Extrapolation — outside the data range**

○ Correlation — measuring association

○ Causation — tutoring causes gains

Question 7

Using y = 4x + 8 from Q6, which is the most reliable prediction?

○ x = 0 hours (no data there)

✓ **x = 4 hours (within the data range)**

○ x = 20 hours (beyond data)

○ x = −2 hours (impossible)

Question 8

A researcher finds y = 50x + 200 models home price (y, in thousands $) based on square footage (x, in hundreds sq ft). What is the predicted price for a 1,500 sq ft home (x = 15)?

✓ **$950,000**

○ $200,000

○ $750,000

○ $950

🛍 Homework Two-Way Table

120 8th grade students surveyed on sport preference split by **lunch period**:

	Basketball 🏀	Soccer ⚽	Other 🏏	Total
1st Lunch	24	**18**	18	**60**
2nd Lunch	30	12	18	**60**
Total	**54**	**30**	**36**	**120**

Question 1

What is the joint frequency of 1st lunch students who prefer Soccer?

○ 30

✓ **18**

○ 12

○ 60

Question 2

What is the marginal frequency for Basketball?

- ○ 24
- ○ 30
- ✓ **54**
- ○ 120

Question 3

What percent of 1st lunch students prefer Basketball?

- ✓ **40%**
- ○ 45%
- ○ 20%
- ○ 50%

Question 4

What percent of 2nd lunch students prefer Basketball?

- ○ 40%
- ✓ **50%**
- ○ 45%
- ○ 25%

Question 5

Based on the basketball percentages (40% vs 50%), is there an association?

- ○ No — the totals are both 60
- ✓ **Yes — 2nd lunch students prefer basketball more (50% vs 40%)**
- ○ No — 40 and 50 are close enough
- ○ Yes — but only if we add more data

Question 6

What is the relative frequency of students who prefer Soccer (out of 120)?

- ✓ **25%**
- ○ 30%
- ○ 15%
- ○ 18%

Question 7

Which cell is highlighted in green? What does that value represent?

- ○ 1st Lunch × Basketball = 24 (most popular combo for 1st lunch)
- ✓ **1st Lunch × Soccer = 18 (joint frequency)**
- ○ Soccer column total = 30
- ○ Grand total = 120

Question 8

If a student is chosen randomly from the 120 surveyed, what is the probability they prefer "Other" sports?

- ○ 18/60 = 30%
- ○ 36/60 = 60%
- ✓ **36/120 = 30%**
- ○ 18/120 = 15%

Extended thinking: These questions require connecting multiple concepts. Take your time and show your reasoning.

Question 1

The line y = −3.5x + 75 models temperature (°F) at altitude x (in 1,000s of feet). At 8,000 ft (x = 8), predict the temperature.

- ✓ **47°F**
- ○ 75°F
- ○ 28°F
- ○ 47°C

Question 2

Using the same model (y = −3.5x + 75): at what altitude does the model predict 0°F? Round to the nearest 1,000 ft.

- ✓ **x ≈ 21,000 ft**
- ○ x ≈ 75,000 ft
- ○ x ≈ 15,000 ft
- ○ x ≈ 10,000 ft

Question 3

Two researchers study "firefighters at scene vs. fire damage." They find a strong positive correlation. Researcher A says: "More firefighters cause MORE damage." Researcher B says: "More firefighters are sent to BIGGER fires, which also cause more damage." Who is correct?

- ○ Researcher A — the data shows clear causation
- ✓ **Researcher B — larger fires need more firefighters AND cause more damage (confounding variable)**
- ○ Both are wrong — there is no correlation
- ○ Both are correct — it depends on the day

Question 4

A two-way table shows: 80% of athletes who do strength training pass fitness tests, vs. 40% of athletes who don't. The difference is 40 percentage points. What does this suggest?

- ○ There is no association
- ✓ **There is a strong association between strength training and passing fitness tests**
- ○ The sample must be biased
- ○ The difference is too small to matter

Question 5

Line of best fit: y = 12x + 5. If x increases by 3 units, by how much does y increase?

- ○ 12
- ○ 15
- ✓ **36**

○ 5

Question 6

Can we use a scatterplot to PROVE that studying more CAUSES higher test scores? Explain.

○ Yes — a strong positive correlation proves causation

✓ **No — correlation does not prove causation; other factors may be involved**

○ Yes — if the trend line is very accurate

○ No — only if there are more than 100 data points

📝 Worked Examples

Study these step-by-step solutions before attempting the problems above.

Two-way table: 45 boys like pizza out of 120 total

Step 1: Joint relative freq

```
45 ÷ 120 = 0.375 = 37.5%
```

Step 2: Conditional: of 60 boys

```
45 ÷ 60 = 0.75 = 75% of boys like pizza
```

🔍 Common Mistakes

Can you spot what went wrong?

Correlation: points go up-left to down-right

✗ **Student:** `Student says: positive correlation`

✓ **Fix:** WRONG! Downhill = NEGATIVE correlation. Uphill = positive.

💡 *Think of it like a hill: going up = positive, going down = negative.*

⚡ Key Tips for This Unit

✅ Positive: both increase. Negative: one up, other down. None: random.

✅ Line of best fit: approx equal points above and below.

✅ Outlier: far from the pattern. Can distort the trend line.

✅ Joint RF = cell÷grand total. Conditional RF = cell÷row or column total.